Method of Sections

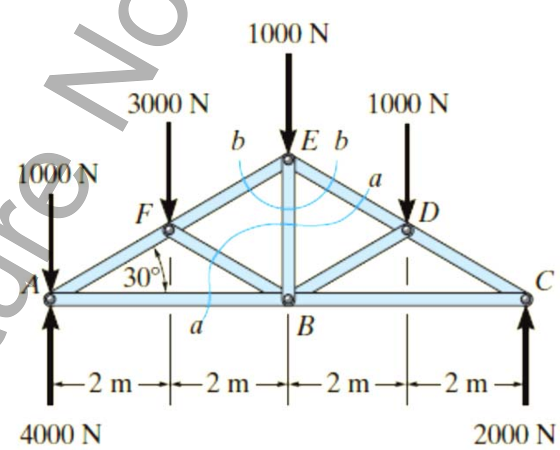
Procedure For Analysis

- **FREE BODY DIAGRAM**
  - Make a decision on how to “cut” or section the truss through the members where forces are to be determined.
  - Before isolating the appropriate section, it may first be necessary to determine the *truss’s support reactions*. If this is done then the three equilibrium equations will be available to solve for member forces at the section.
  - Draw the free-body diagram of that segment of the sectioned truss which has the least number of forces acting on it.
- **EQUATIONS OF EQUILIBRIUM**
  - Moments should be summed about a point that lies at the intersection of the lines of action of two unknown forces, so that the third unknown force can be determined directly from the moment equation.
  - If two of the unknown forces are *parallel*, forces may be summed *perpendicular* to the direction of these unknowns to determine directly the third unknown force.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 282

Loads

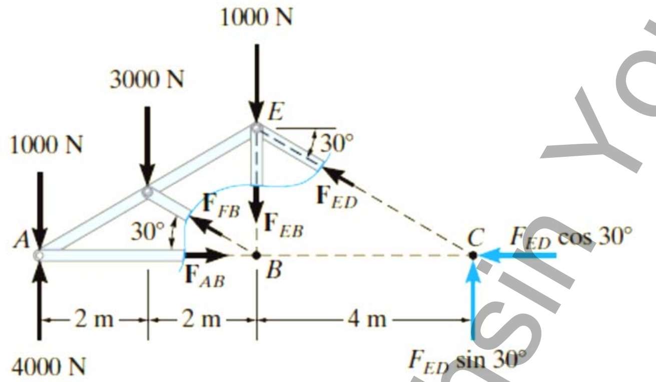
Internal Load in Trusses - Method of Sections



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.7, Fig 6-18a, pp. 285

Loads

Internal Load in Trusses - Method of Sections

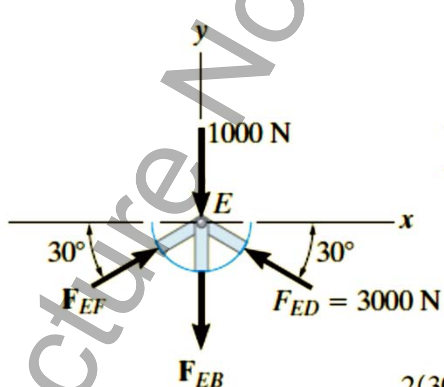


$$\zeta + \Sigma M_B = 0; \quad 1000\text{ N}(4\text{ m}) + 3000\text{ N}(2\text{ m}) - 4000\text{ N}(4\text{ m}) + F_{ED}\sin 30^\circ(4\text{ m}) = 0$$

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.7, Fig 6-18b pp. 285

Loads

Internal Load in Trusses - Method of Sections



$$\xrightarrow{+} \Sigma F_x = 0;$$

$$F_{EF}\cos 30^\circ - 3000\cos 30^\circ\text{ N} = 0$$

$$F_{EF} = 3000\text{ N} \quad (\text{C})$$

$$+\uparrow \Sigma F_y = 0;$$

$$2(3000\sin 30^\circ\text{ N}) - 1000\text{ N} - F_{EB} = 0$$

$$F_{EB} = 2000\text{ N} \quad (\text{T})$$

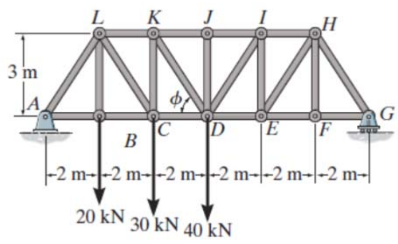
[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.7, Fig 6-18c pp. 285

Method of Sections

FUNDAMENTAL PROBLEMS

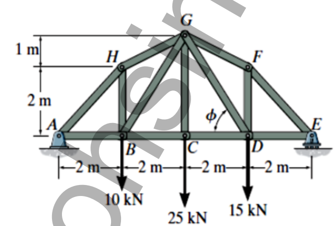
F6–8. Determine the force in members *LK*, *KC*, and *CD* of the Pratt truss. State if the members are in tension or compression.

F6–9. Determine the force in members *KJ*, *KD*, and *CD* of the Pratt truss. State if the members are in tension or compression.



F6–8/9

F6–11. Determine the force in members *GF*, *GD*, and *CD* of the truss. State if the members are in tension or compression.

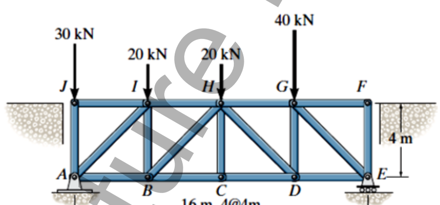


[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 286

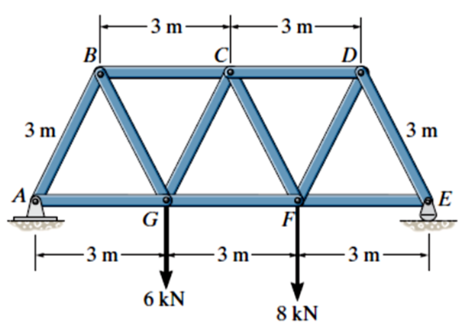
Method of Sections

PROBLEMS

***6–32.** The *Howe bridge truss* is subjected to the loading shown. Determine the force in members *HD*, *CD*, and *GD*, and state if the members are in tension or compression.



•6–37. Determine the force in members *CD*, *CF*, and *FG* of the *Warren truss*. Indicate if the members are in tension or compression.



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 287